

MCS 481 Project Two : locating CTA stops with kd trees
due Wednesday 21 October 2026 at noon

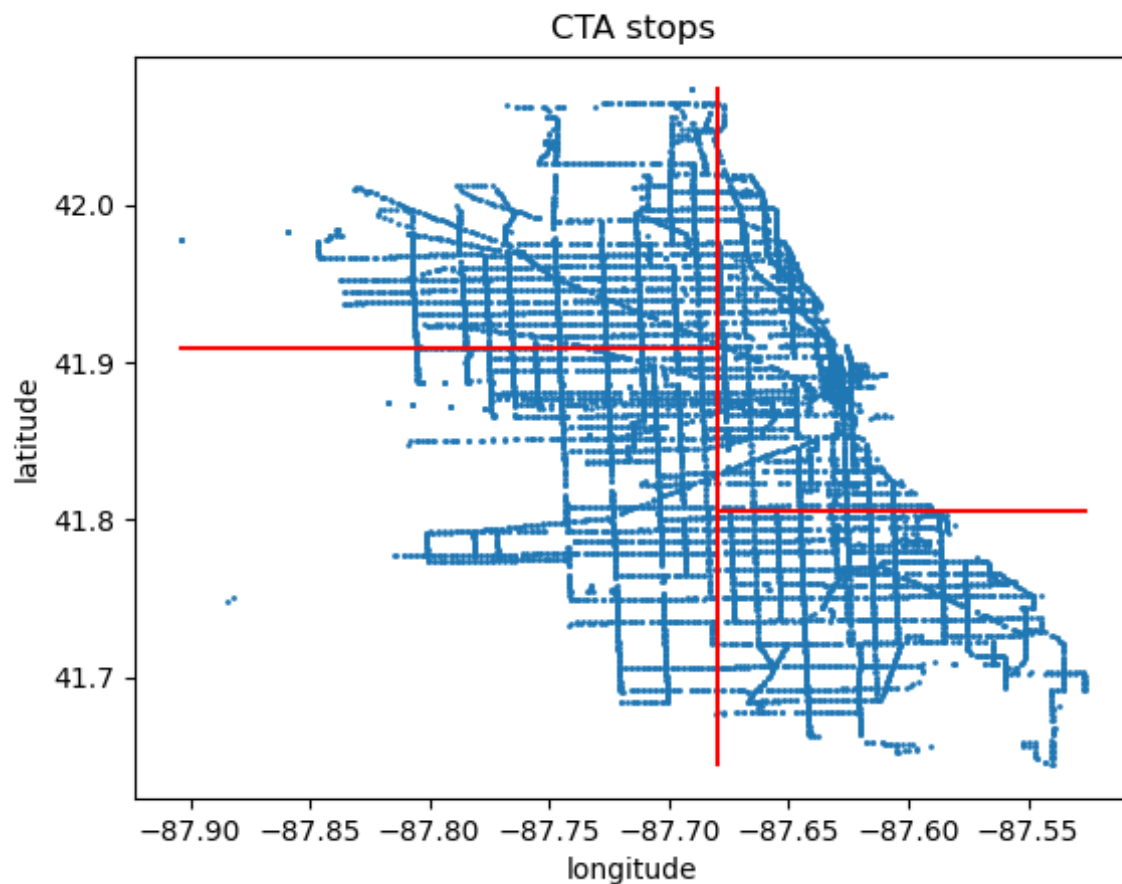
The goal of this second project is to apply kd trees to locate stops of the Chicago Transit Authority (CTA) via queries made on a kd tree.

1. Stops of the Chicago Transit Authority

All coordinates of the CTA stops are available at

https://www.transitchicago.com/downloads/sch_data/

in the file `stops.txt` of the `google_transit` folder. In a comma separated value format, the latitudes and the longitudes appear in `stops.txt` respectively on the fifth and fourth position from the end of each line. Using longitude and latitude as the (x, y) -coordinates of the stops, a basic scatterplot made with matplotlib is shown below. The red lines on the plot are the splitting lines on the medians.



Assignment One. Download the coordinates of all CTA stops and make the above plot.

1. For each quadrant defined by the first three splitting lines, add the splitting lines which split each quadrant once more in four.
2. Construct a kd tree for the points defined by the CTA stops.
Compare the time it takes to construct a kd tree with the time it takes to download and construct the input for the construction of a kd tree.

2. Making Queries

Given a position, specified by longitude and latitude, we would like to know the locations of the CTA stops nearest to the given position.

Assignment Two. Use a kd tree to make queries.

1. Define a function which takes on input a position and a radius representing the distance from the given position for use in the query to locate the nearest CTA stops.
2. The output of your function should contain the list of nearest CTA stops and the time it took to execute the query.
3. Compare the times for small and large values of the radius.
Are the times for the queries output sensitive?

3. Smallest and Largest Empty Query Intervals

The basic scatterplot of CTA plots shows some relatively large empty rectangles, rectangles for which a query would report no CTA stops, which could represent underserved neighborhoods of the city.

Assignment Three. Design a systematic strategy to look for the largest empty query rectangles, excluding the obvious lake, and the farthest suburbs.

4. the Deadline is Wednesday 21 October 2026 at noon

Upload your answers to gradescope before noon on Wednesday 21 October 2026.

Your solution consists of

1. The specification of your computers and the version of the software used.
2. The code for the programs you used in the experiments.
3. All relevant plots, preferably embedded in the output cells of the submitted Jupyter notebook, or otherwise uploaded as separate png files.
4. Tables with properly formatting of the essential characteristics for each experiment: the dimension of input and output, and the running time. Raw data may be provided as an appendix.
5. Your interpretation of the results and conclusions.

You may work in pairs for this project.

If you work in a pair, then one member of the pair must email the name of the partner to janv@uic.edu before 5pm on Wednesday 14 October.

If you have questions or difficulties with the project, feel free to ask questions during office hour.